

3. Temperature

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Pre-class materials

i Read ahead

Before class, you can prepare by reading the following materials:

1. Withers chapter 5. Look over the homework, handout, and discussion questions and skim the chapter, reading anything that you need to know.
2. [\[Discussion Questions\]](#)
3. [\[Slide Deck\]](#) - for your reference as we go through the material

Announcements/Reminders

- Labs this week - Meet at Honolulu Zoo at 1:30pm. LMK if you need a ride from campus. [\[Read Lab 2\]](#). Bring:
- Your lab notebook
- A timing device (e.g. your phone)
- Sun screen, hat, and sunglasses. It will be hot and sunny!
- Water
- One hardcopy of Lab report to turn in
- Next week we will do the [\[Toad Heart Lab\]](#).
- We will finish Metabolism and begin Temperature.
- HW1 due Tuesday in class, hand-written (you may submit as a group - week 2 if you wish).
- HW2 due next Tuesday in class (individual).

Week 3 Discussion Groups

Group	Partner 1	Partner 2	Partner 3
1	Kyra	Dallas	Trey
2	Niki	Carla	Xavier
3	Peigi	Alice	Manuel
4	Reagan	Emma	
5	Paris	Darwin	

Thought for the day

Physiology is the story of evolution's struggle to maintain an appropriate SA/D ratio in relation to the volume of an animal – *Haldane*

3. Temperature

Modes of Heat Transfer

- Conduction
- Convection
- Evaporative
- Radiative

Animals also generate heat through **Metabolism**.

<https://youtu.be/BHchDrboqEo>

Flux = C * ∇ (Mass or Energy)

Q = C * M ∇ T

Where: - Flux is the transfer of mass or energy - ∇ is the gradient symbol - Q is heat - C is a material property (resistance or insulation value) - T is temperature

Heat Balance

Metabolic heat production is balanced by all mechanisms for heat exchange:

$$\Delta H_s = H_m \pm H_c \pm H_r \pm H_e$$

Where:

ΔH_s = heat of storage

H_m = heat of metabolism

H_c = heat of conduction and convection

H_r = heat of radiation

H_e = heat of evaporation

If an animal is in heat balance, T_b (body temperature) is stable.

If ΔH_s is positive, (Gains > Losses), then T_b increases.

If ΔH_s is negative, (Gains < Losses), then T_b decreases.

Body Temperature

Q10

https://youtu.be/T5O9UvSZ_-g

Iterative Method

<https://youtu.be/pEzcZCTYPyE>

For Next Time

1. [HW2] due on Tuesday in class. (Individual)
2. Finish temperature, begin [Cardiac Function]
3. Start working on design 1 calculations with your partner. See me for help [Office hours]
4. Next Lab: [Toad Heart Lab]
5. Please fill out the TEAMMATES for both discussion week 3 and Lab 2 group lab by WEDNESDAY. (check your email)